# Practice workbook · Draw or label a human cell as seen under the microscope

Labelling given diagrams of human cells, naming lettered parts, naming a missing organelle, or drawing a labelled cheek cell.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DRAW-LABEL-ANIMAL-CELL

# 1. Question 2 · part a · 4 marks

November 2023 Paper 2

Worked-example retry · Attempt part a only.

**2** (a) Draw a labelled diagram of a cheek cell from inside a mouth as seen under a light microscope.

**(4)**

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(b) (i) Explain what is meant by the term **tissue**.

**(3)**

(ii) State the type of tissue that contains cheek cells.

**(1)**

(iii) Name two other types of tissue.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 10 marks)**

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## 1. Question 2 · part a · 4 marks · Working and answer

Working and answer

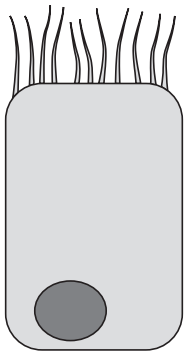
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# 2. Question 2 · part a.i · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

**2** The diagram shows a cell from the human breathing system.



(a) (i) Add these labels to the diagram.

- cilia
- cell membrane
- nucleus

**(3)**

(ii) Each of the cell structures has a particular function.

The boxes give a list of structures and functions.

Draw one straight line from each structure to its function.

**(3)**

| Structure | Function |
|---|---|
| cilia | holds the genetic code |
| cell membrane | site where most chemical reactions occur |
| nucleus | controls the entry and exit of substances |
| | makes proteins |
| | moves mucus through the trachea |

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## 2. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

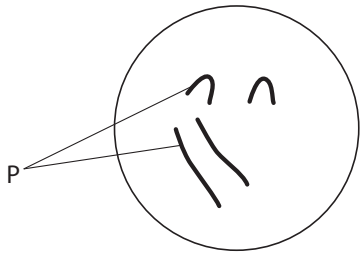
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## 3. Question 8 · part a.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part a.i only.

**8** (a) The diagram shows the nucleus of a cell.




(i) Name the structures labelled P.

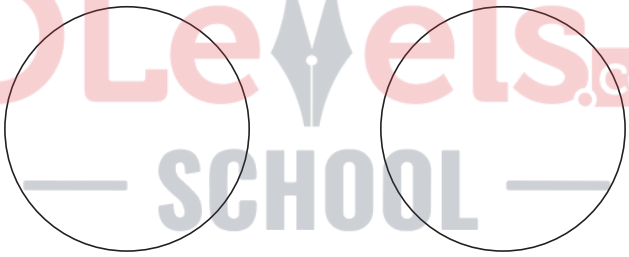
**(1)**

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(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

**(2)**



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

**(1)**

| Stage | Number |
|---|---|
| complementary bases pair up | |
| DNA double helix unwinds | |
| strands separate | |
| two DNA helices form | |

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P 6 2 0 5 1 A 0 2 2 2 4

## 3. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

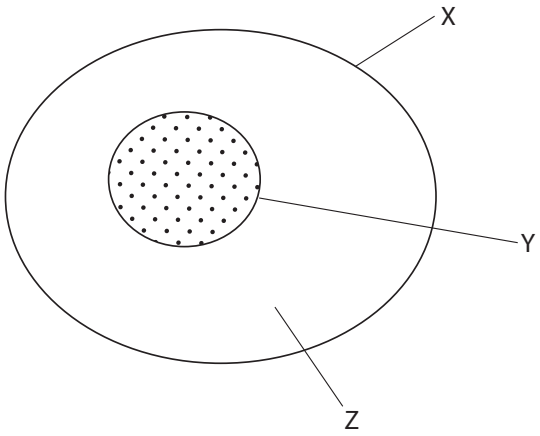
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# 4. Question 1 · part a.i · 3 marks

June 2022 Paper 1R

Further practice · Attempt part a.i only.

**1** (a) The diagram shows a human body cell.




(i) Name the parts X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

**(1)**

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(iii) Explain how body cells are organised to form organs.

**(2)**

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## 4. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

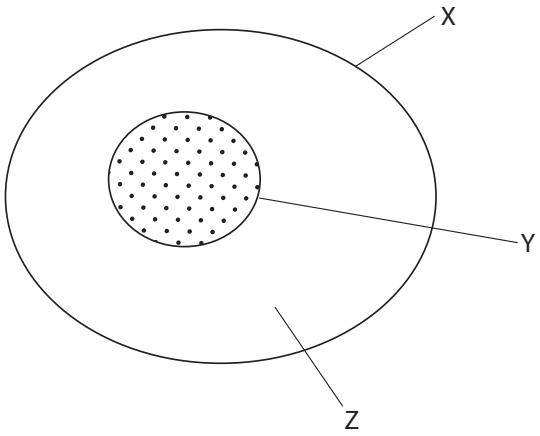
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# 5. Question 1 · part a.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

**1** (a) The diagram shows a human body cell.




(i) Name the parts X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

**(1)**

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(iii) Explain how body cells are organised to form organs.

**(2)**

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## 5. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

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# 6. Question 6 · part a · 4 marks

June 2023 Paper 1R

Further practice · Attempt part a only.

**6** The lining of the inside of the cheek is made of squamous epithelial cells.

(a) Draw a labelled diagram to show a cheek cell when seen using a light microscope.

**(4)**

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(b) Describe how a sample of cheek cells would be prepared to view using a light microscope.

**(3)**



P 7 2 5 8 6 A 0 1 4 2 0

## 6. Question 6 · part a · 4 marks · Working and answer

Working and answer

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